

Basic ray theory

Lagrange-Hamilton theory

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Introduction

- Seismic ray theory:
 - Solution to the elastic wave equation by the «method of characteristics»
 - High-frequency approximation
- Wave theory vs. ray theory
 - Wave theory: Heisenberg's uncertainty relation applies; x or p can be known exactly (but not both)
 - Ray theory: Heisenberg is ignored; both x and p can be known exactly.

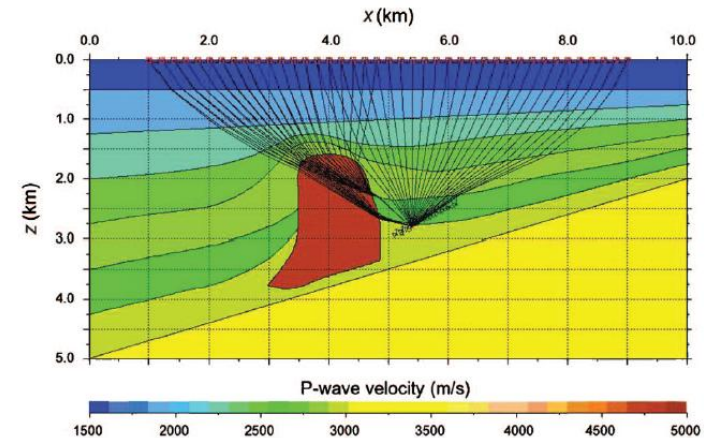


Figure 6. Synthetic velocity model with an embedded salt body and a simulated deviated well. Rays traced from level number nine to all surface sampling locations are indicated.

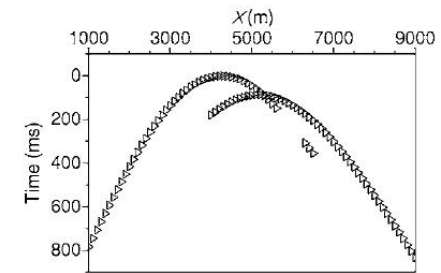


Figure 7. Simulated direct ray traveltime data. The data correspond to a drill-bit source at level number nine.

Acoustic wave equation for displacement

Newton and Hooke in acoustic medium:

$$\begin{aligned}\rho \partial_t^2 u_i &= \partial_i \sigma + f_i \\ \sigma &= M \partial_j u_j\end{aligned}$$

Wave equation for displacement:

$$\rho \partial_t^2 u_i = \partial_i (M \partial_j u_j) + f_i$$

Assume $f_i = 0$, $\partial_j \rho \simeq 0$ and $\partial_j M \simeq 0$ (smooth medium):

$$\begin{aligned}\frac{1}{v^2} \partial_t^2 u_i - \partial_i \partial_j u_j &= 0 \\ v &= \sqrt{M/\rho}\end{aligned}$$

Eikonal equation (1)

Ray ansatz:

$$u_i = A(\mathbf{x})g_j e^{-i\omega(t-T)}$$

Derivatives:

$$\partial_t^2 u_i = -\omega^2 u_i$$

$$\partial_j u_j = [\partial_j (Ag_j) + Ag_j (i\omega \partial_j T)] e^{-i\omega(t-T)}$$

$$\begin{aligned} \partial_i \partial_j u_j &= [\partial_i \partial_j (Ag_j) + \partial_j (Ag_j) (i\omega \partial_i T) \\ &+ \partial_i (Ag_j) (i\omega \partial_j T) + (Ag_j) (i\omega \partial_i \partial_j T) \\ &- Ag_j \omega^2 \partial_j T \partial_i T] e^{-i\omega(t-T)} \end{aligned}$$

$$\partial_i \partial_j u_j = [f_0 + i\omega f_1 - \omega^2 Ag_j \partial_j T \partial_i T] e^{-i\omega(t-T)}$$

Eikonal equation (2)

Substitute derivatives in wave equation:

$$(Ag_i \frac{\omega^2}{v^2} + [f_0 + i\omega f_1 - \omega^2 Ag_j \partial_j T \partial_i T])e^{-i\omega(t-T)} = 0$$

Divide by ω^2 , and let $\omega \rightarrow \infty$:

$$\lim_{\omega \rightarrow \infty} \left(Ag_i \frac{1}{v^2} - \left[\frac{f_0}{\omega^2} + \frac{if_1}{\omega} + Ag_j \partial_j T \partial_i T \right] \right) = 0$$

Finally, use normalization $g_i g_i = 1$ and $\nabla T \parallel \mathbf{g}$ (P-waves):

$$\left(g_i \frac{1}{v^2} - [g_j \partial_j T \partial_i T] \right) g_i = 0$$

$$\text{Eikonal equation: } \frac{1}{v^2} - |\nabla T|^2 = 0$$

Lagrange-Hamilton theory

- Alternative formulation of mechanics (higher level of abstraction)
- Focused on energy rather than forces

Lagrangian:

$$L = L(x_i, \dot{x}_i, t) = K - U \quad (i \in \{1, 2, 3, \dots, 3n\})$$

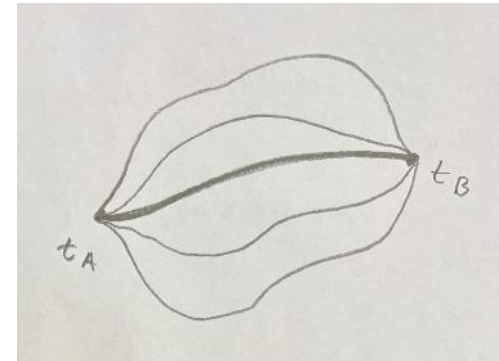
K = kinetic energy

U = potential energy

Principle of least action:

$$S = \int_{t_A}^{t_B} L(x_i, \dot{x}_i, t) dt$$

$$\Delta S = 0 \quad \text{for the physical path}$$



Reference: Goldstein, H., 1951, Classical Mechanics

Euler-Lagrange equations

Variation of the action:

$$\Delta S = \int_{t_A}^{t_B} \left[\frac{\partial L}{\partial x_i} \Delta x_i + \frac{\partial L}{\partial \dot{x}_i} \Delta \dot{x}_i \right] dt = 0$$

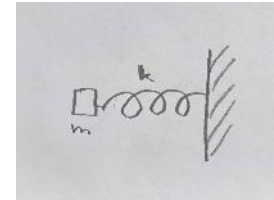
Integration by parts:

$$\Delta S = \int_{t_A}^{t_B} \left[\frac{\partial L}{\partial x_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} \right] \Delta x_i dt + \cancel{\frac{\partial L}{\partial \dot{x}_i} \Delta x_i} \Big|_{t_A}^{t_B} = 0$$

Euler-Lagrange equations:

$$\frac{\partial L}{\partial x_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} = 0$$

Example:
Harmonic oscillator



$$K = \frac{1}{2} m \dot{x}^2$$

$$U = \frac{1}{2} k x^2$$

$$\frac{\partial L}{\partial x} = -kx$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = m\ddot{x}$$

$$m\ddot{x} = -kx$$

Hamiltonian

Define momentum

$$p_i = \frac{\partial L}{\partial \dot{x}_i}$$

Then, using Euler-Lagrange:

$$\dot{p}_i = \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} = \frac{\partial L}{\partial x_i}$$

Define Hamiltonian (Legendre transform):

$$H = p_i \dot{x}_i - L$$

- 6D phase space: $\{x_i, p_i\}$

Example:
Harmonic oscillator

$$K = \frac{1}{2} m \dot{x}^2$$

$$U = \frac{1}{2} k x^2$$

$$p = \frac{\partial L}{\partial \dot{x}} = m \dot{x}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}$$

$$\dot{p} = m \ddot{x} = -kx$$

$$\begin{aligned} H &= p \dot{x} - L \\ &= m \dot{x}^2 - (K - U) \\ &= K + U \end{aligned}$$

Differential, from definition of H:

$$\begin{aligned}
 \Delta H &= \dot{x}_i \Delta p_i + p_i \Delta \dot{x}_i - \Delta L \\
 &= \dot{x}_i \Delta p_i + p_i \Delta \dot{x}_i - \left[\frac{\partial L}{\partial x_i} \Delta x_i + \frac{\partial L}{\partial \dot{x}_i} \Delta \dot{x}_i + \frac{\partial L}{\partial t} \Delta t \right] \\
 &= \dot{x}_i \Delta p_i + \left[p_i - \cancel{\frac{\partial L}{\partial \dot{x}_i}} \right] \Delta \dot{x}_i - \frac{\partial L}{\partial x_i} \Delta x_i - \frac{\partial L}{\partial t} \Delta t \\
 &= \dot{x}_i \Delta p_i - \dot{p}_i \Delta x_i - \frac{\partial L}{\partial t} \Delta t
 \end{aligned}$$

Also:

$$\Delta H = \frac{\partial H}{\partial p_i} \Delta p_i + \frac{\partial H}{\partial x_i} \Delta x_i + \frac{\partial H}{\partial t} \Delta t$$

Hamiltons equations:

$$\begin{aligned}
 \dot{x}_i &= \frac{\partial H}{\partial p_i} \\
 \dot{p}_i &= -\frac{\partial H}{\partial x_i}
 \end{aligned}$$

$$\frac{\partial H}{\partial t} = \frac{dH}{dt} = -\frac{\partial L}{\partial t}$$

Time derivative of Hamiltonian

$$\begin{aligned}\frac{dH}{dt} &= \dot{x}_i \dot{p}_i + \ddot{x}_i p_i - \frac{dL}{dt} \\&= \dot{x}_i \dot{p}_i + \ddot{x}_i p_i - \left[\frac{\partial L}{\partial x_i} \dot{x}_i + \frac{\partial L}{\partial \dot{x}_i} \ddot{x}_i + \frac{\partial L}{\partial t} \right] \\&= \dot{x}_i \left(\cancel{\dot{p}_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} \right) + \ddot{x}_i \left(\cancel{p_i} - \frac{\partial L}{\partial \dot{x}_i} \right) - \frac{\partial L}{\partial t} \\&= -\frac{\partial L}{\partial t}\end{aligned}$$

Emmy Noether (1882-1935)



Why do we have
conservation laws?

Invariance of the Lagrangian (action)

Continues transformations

$$\begin{aligned}x_i &\rightarrow x'_i = x_i + \eta_i \\ t &\rightarrow t' = t + \tau\end{aligned}$$

Invariance of the Lagrangian:

$$L(x_i, t) = L(x'_i, t')$$



$$L(x_i, t) - L(x'_i, t') = \frac{\partial L}{\partial x_i} \eta_i + \frac{\partial L}{\partial t} \tau = 0$$

Noether's theorem

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} \eta_i - \frac{dH}{dt} \tau = 0$$

$$\frac{d}{dt} [p_i \eta_i - H \tau] = 0$$

For every continuous symmetry transformation of the Lagrangian (action), there is a conservation law

Time translation symmetry =>
Conservation of energy

$$\begin{aligned} \eta_i &= 0 \\ \tau &> 0 \\ \tau \frac{dH}{dt} &= 0 \end{aligned}$$

Spatial translation symmetry =>
Conservation of momentum

$$\begin{aligned} \eta_1 &> 0 \\ \eta_2 &= \eta_3 = 0 \\ \tau &= 0 \\ \eta_1 \frac{p_i}{dt} &= 0 \end{aligned}$$

Noether's theorem

- For every continuous symmetry transformation of the Lagrangian (action), there is a conservation law

- The converse is possibly not true (?)

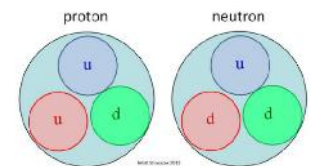
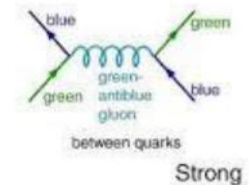
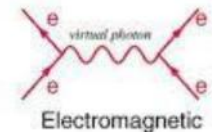
Examples:

- Time translation => conservation of energy
- Spatial translation => conservation of momentum (slowness)
- Rotation => conservation of angular momentum
- Spin
- Abelian gauge symmetry => conservation of charge
- Non-Abelian gauge symmetry => conservation of color (QCD)

Continuum and fields:

- Lagrangian density

$$S = \int \mathcal{L} d^3x dt$$
$$L = \int \mathcal{L} d^3x$$



Seismic ray theory.

Method of characteristics; Hamiltonian

Find $x_i = x_i(u)$ and $p_i(u)$ such that

$$\mathcal{H}(p_i, x_i) = 0, \quad (i = 1, 2, 3)$$

(p_i, x_i) are coordinates in 6D **phase space**.

u is a monotonically increasing parameter, usually one of

$$u = T, \text{ (traveltime)}$$

$$u = s, \text{ (arc length)}$$

Here we will use $u = T$; convenient in anisotropic media.

Hill (1990) uses $u = s$.

Different forms of the Hamiltonian

The Hamiltonian $\mathcal{H}$ can be defined in many ways:

$$\mathcal{H}_0 = v^2 p_i p_i - 1 = 0$$

$$\mathcal{H}_1 = |\mathbf{p}| - \frac{1}{v} = 0$$

$$\mathcal{H}_2 = p_i p_i - \frac{1}{v^2} = 0$$

All are OK; motivated by the eikonal equation.

Hamilton canonical equations (1)

Complete differential:

$$d\mathcal{H} = \frac{\partial \mathcal{H}}{\partial p_i} dp_i + \frac{\partial \mathcal{H}}{\partial x_i} dx_i = 0$$

$$\frac{\partial \mathcal{H}}{\partial p_i} dp_i = -\frac{\partial \mathcal{H}}{\partial x_i} dx_i$$

which is fulfilled if

$$\begin{aligned} dp_i &= \frac{\partial \mathcal{H}}{\partial x_i} du \\ dx_i &= -\frac{\partial \mathcal{H}}{\partial p_i} du \end{aligned}$$

where u is any monotonically increasing parameter.

Hamilton canonical equations (2)

Hamilton equations

$$\begin{aligned}\frac{dp_i}{du} &= \frac{\partial \mathcal{H}}{\partial x_i} \\ \frac{dx_i}{du} &= -\frac{\partial \mathcal{H}}{\partial p_i}\end{aligned}$$

Let $u = T$ (traveltime along the ray):

$$\begin{aligned}\frac{dp_i}{dT} &= \frac{\partial \mathcal{H}}{\partial x_i} \\ \frac{dx_i}{dT} &= -\frac{\partial \mathcal{H}}{\partial p_i}\end{aligned}$$

- The Hamilton equations are of fundamental importance in theoretical physics:
 - Classical mechanics
 - Classical field theory
 - Quantum field theory
 - Ray theory
- Some good books:
 - Landau and Lifshitz (1939): The Classical Theory of Fields
 - Goldstein (1951): Classical Mechanics
 - Morse and Feshbach (1953): Methods of Theoretical Physics.
 - Cerveny (2001): Seismic Ray Theory

Equation for T

Sometimes (if $u = s$) we need an equation for T

$$\begin{aligned}\frac{dT}{du} &= \frac{\partial T}{\partial x_i} \frac{dx_i}{du} \\ &= p_i \frac{dx_i}{du} \\ &= -p_i \frac{\partial \mathcal{H}}{\partial p_i}\end{aligned}$$

Choice of u vs Hamiltonian:

$$\begin{aligned}u = s &\Rightarrow \frac{dT}{du} = \frac{dT}{ds} = \frac{1}{v} \Rightarrow \mathcal{H} = \mathcal{H}_1 \text{ (Hill)} \\ u = T &\Rightarrow \frac{dT}{du} = \frac{dT}{dT} = 1 \Rightarrow \mathcal{H} = \mathcal{H}_0\end{aligned}$$

Anisotropic elastic wave equation

Newton and Hooke:

$$\begin{aligned}\rho \partial_t^2 u_i &= \partial_j \sigma_{ij} \\ \sigma_{ij} &= c_{ijkl} \partial_k u_l\end{aligned}$$

Wave equation:

$$\rho \partial_t^2 u_i = \partial_j c_{ijkl} \partial_k u_l$$

Smooth medium ($\partial_j c_{ijkl} \simeq 0$):

$$\partial_t^2 u_i = \frac{c_{ijkl}}{\rho} \partial_j \partial_k u_l$$

Anisotropic elastic ray theory

Ray ansatz:

$$u_i(\mathbf{x}, t) = Ag_j e^{-i\omega(t-T)}$$

Substitute in wave equation and let $\omega \rightarrow \infty$:

$$\frac{c_{ijkl}}{\rho} p_j p_l g_k - g_i = 0$$

Christoffel equation:

$$[\Gamma_{ik} - \delta_{ik}]g_k = 0$$

Christoffel tensor:

$$\Gamma_{ik} = \frac{c_{ijkl}}{\rho} p_j p_l$$

Properties of the Christoffel tensor:

$$\begin{aligned}\Gamma_{ik} &= \Gamma_{ki} \text{ (symmetric)} \\ \Gamma_{ik}(a\mathbf{p}) &= a^2 \Gamma_{ik}(\mathbf{p})\end{aligned}$$

Homogeneous function of 2nd degree

Eigenvalues and eigenvectors

Eigenvalues:

$$\begin{aligned}\det[\mathbf{\Gamma} - G^{(m)}\mathbf{I}] &= 0 \\ G^{(m)}(\mathbf{p}) &= 1\end{aligned}$$

Eigenvectors:

$$[\mathbf{\Gamma} - G^{(m)}\mathbf{I}]\mathbf{g}^{(m)} = 0 \qquad [\Gamma_{ik} - G^{(m)}\delta_{ik}]g_k^{(m)} = 0$$

Normalization:

$$\mathbf{g}^{(m)T}\mathbf{g}^{(m)} = 1 \qquad g_i^{(m)}g_i^{(m)} = 1$$

3 eigenvalues, 3 eigenvectors $\Rightarrow$ 3 wave modes:

$$m = \begin{cases} 1 & : \text{qS1 waves} \\ 2 & : \text{qS2 waves} \\ 3 & : \text{qP waves} \end{cases}$$

More about the eigenvalues

Multiply by $g_i^{(m)}$ and sum over i, j :

$$[\Gamma_{ik} - G^{(m)}\delta_{ik}]g_k^{(m)}g_i^{(m)} = 0$$

Explicit equation for eigenvalues:

$$G^{(m)} = \Gamma_{ik}g_k^{(m)}g_i^{(m)}$$

Using the homogeneous function property:

$$G^{(m)}(\mathbf{p}) = 1 \quad (\text{to fulfill Christoffel equation})$$

$$G^{(m)}(\mathbf{n}) = v^2 \quad (\text{by definition of phase normal})$$

Slowness and phase direction:

$$\mathbf{p} = \nabla T = \frac{\mathbf{n}}{v}$$

$$v = \text{phase velocity, } \mathbf{n} = \text{phase normal, } |\mathbf{n}| = 1$$

Hamiltonian and Hamilton equations

Hamiltonian:

$$\mathcal{H} = \frac{1}{2}(G^{(m)} - 1) = 0$$

$$\mathcal{H} = \mathcal{H}_0$$

Hamilton equations:

$$\begin{aligned}\frac{\partial x_i}{\partial T} &= \frac{\partial \mathcal{H}}{\partial p_i} = \frac{1}{2} \frac{\partial G^{(m)}}{\partial p_i} \\ \frac{\partial p_i}{\partial T} &= -\frac{\partial \mathcal{H}}{\partial x_i} = -\frac{1}{2} \frac{\partial G^{(m)}}{\partial x_i}\end{aligned}$$

Kinetic ray tracing system,
6 first-order PDEs, to be
solved simultaneously

Group velocity (ray velocity):

$$V = \left| \frac{d\mathbf{x}}{dT} \right|$$

$$V = \sqrt{V_i V_i} = \sqrt{\frac{dx_i}{dT} \frac{dx_i}{dT}}$$

The group velocity vector is tangent to the ray

Dynamic ray tracing, geometrical spreading

Ray ansatz:

$$u_i(\mathbf{x}, t) = Ag_j e^{-i\omega(t-T)}$$

Ray amplitude:

$$A(\mathbf{x}, \mathbf{x}_0) = \frac{\mathcal{R}^c(\mathbf{x}, \mathbf{x}_0) e^{i\frac{\pi}{2} \text{sgn}(\omega)\sigma}}{4\pi Z(\mathbf{x}) Z(\mathbf{x}_0) \mathcal{L}(\mathbf{x}, \mathbf{x}_0)}$$

σ = KMAH index

$$Z(\mathbf{x}) = \rho(\mathbf{x}) v(\mathbf{x})$$

v = phase velocity

$$\mathcal{R}^c(\mathbf{x}, \mathbf{x}_0) = \prod_k R_k \left[\frac{(\rho V \cos \psi)_{k+}}{(\rho V \cos \psi)_{k-}} \right]^{1/2}$$

V = group velocity

ψ = group angle

$$\mathcal{L}(\mathbf{x}, \mathbf{x}_0) = [\det \mathbf{Q}_2]^{1/2} = \left[\frac{\det \mathbf{Q}}{v} \right]^{1/2}$$

What is $\mathbf{Q}_2$ and $\mathbf{Q}$?

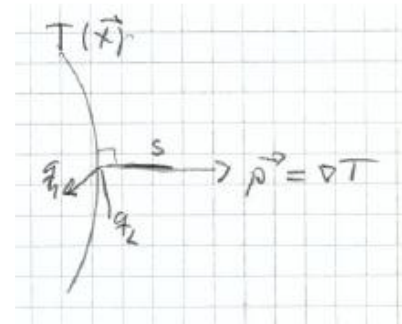
Local phase and group (ray) coordinates

Local ray-centered phase coordinates:

$$\mathbf{q} = (q_1, q_2, s)$$

$$\hat{\mathbf{s}} \parallel \mathbf{p}$$

$$\mathbf{p} = (0, 0, p) = (0, 0, 1/v)$$

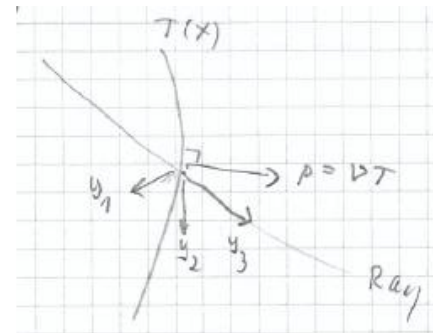


Local ray-centered group coordinates:

$$\mathbf{y} = (y_1, y_2, y_3)$$

$$y_1 = y_2 = 0 \text{ on the ray}$$

$$\mathbf{V} = (0, 0, V)$$



Isotropic media:

$$(q_1, q_2, s) = (y_1, y_2, y_3)$$

Relative geometrical spreading

In ray-centered coordinates:

$$\mathcal{L}(\mathbf{x}, \mathbf{x}_0) = [\det \mathbf{Q}_2]^{1/2}, \quad (2 \times 2 \text{ matrix})$$

$$(Q_2)_{IJ} = \frac{\partial^2 T}{\partial q_I \partial q_J}, \quad (I, J = 1, 2)$$

In Cartesian coordinates:

$$\mathcal{L}(\mathbf{x}, \mathbf{x}_0) = \left[\frac{\det \mathbf{Q}}{v} \right]^{1/2} \quad (3 \times 3 \text{ matrix})$$

$$Q_{iJ} = \frac{\partial x_i}{\partial \gamma_j} \quad \gamma = (\theta, \phi) = \text{phase direction}$$

$$Q_{i3} = \frac{\partial x_i}{\partial T} = V_i$$

Q_{IJ} , ($J = 1, 2$) are computed by dynamic ray tracing

Dynamic ray tracing

3×2 matrices $\mathbf{Q}$ and $\mathbf{P}$:

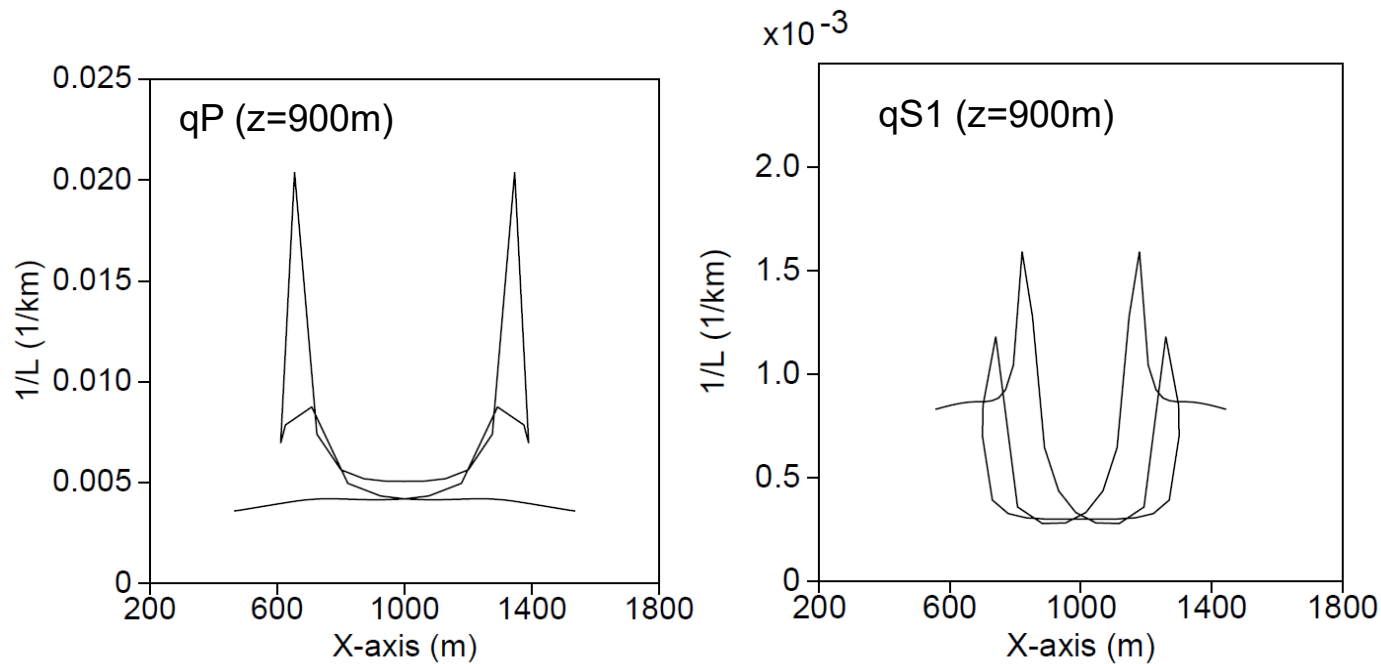
$$Q_{iK} = \frac{\partial x_i}{\gamma_K}$$
$$P_{iK} = \frac{\partial p_i}{\gamma_K}$$

$\gamma = (\theta, \phi) = \text{phase direction}$

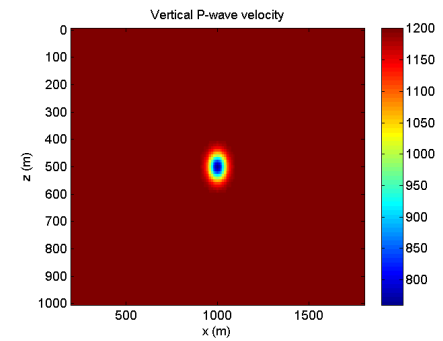
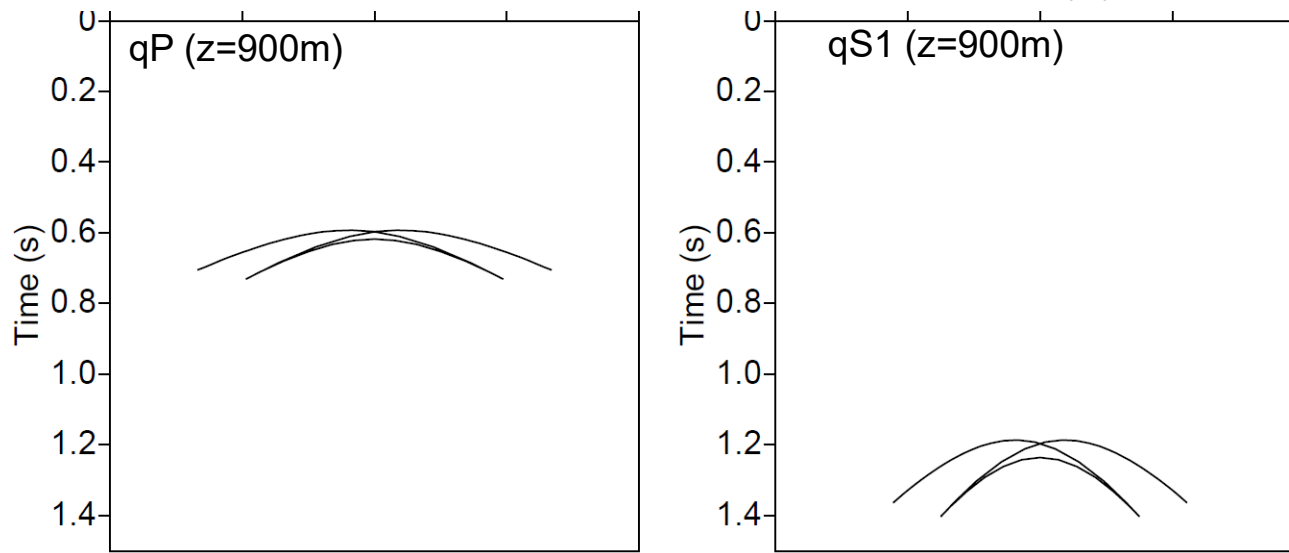
Dynamic ray tracing system:

$$\frac{\partial Q_{iK}}{\partial T} = \frac{\partial^2 \mathcal{H}}{\partial p_i \partial x_j} Q_{jK} + \frac{\partial^2 \mathcal{H}}{\partial p_i \partial p_j} P_{jK}$$
$$\frac{\partial P_{iK}}{\partial T} = -\frac{\partial^2 \mathcal{H}}{\partial x_i \partial x_j} Q_{jK} - \frac{\partial^2 \mathcal{H}}{\partial x_i \partial p_j} P_{jK}$$

Dynamic ray tracing system; 12 first-order PDEs, to be solved simultaneously



Geometrical spreading



Traveltime

Seismograms from ray racing

Sum over events and FFT from frequency to time:

$$u(\mathbf{x}, t) = \int d\omega \sum_n A_n(\mathbf{x}, \mathbf{x}_s) e^{-i\omega[t-T(\mathbf{x}, \mathbf{x}_s)]}$$
$$A_n(\mathbf{x}, \mathbf{x}_0) = \frac{\mathcal{R}^c(\mathbf{x}, \mathbf{x}_0) e^{i\frac{\pi}{2} \text{sgn}(\omega)\sigma}}{4\pi Z(\mathbf{x}) Z(\mathbf{x}_0) \mathcal{L}(\mathbf{x}, \mathbf{x}_0)}$$

- Trace events one by one
- Event specification by ray codes

Common-shot Kirchhoff migration revisited

Substitute U and D in imaging condition:

$$I(x; x_s) = \int d\omega \int \int dx_r dy_r p_z(x_r) \frac{\mathcal{L}(x, x_s)}{\mathcal{L}(x, x_r)} e^{-i\omega[T(x, x_r) + T(x, x_s)]} i\omega P(x_r, \omega)$$

The integral over ω is the inverse Fourier transform:

$$I(x; x_s) = -2 \int \int dx_r dy_r W(x, x_r, x_s) \partial_t P(x_r, \tau)$$

The integral over ω is the inverse Fourier transform:

$$\begin{aligned} \tau &= T(x, x_r) + T(x, x_s) = \text{total traveltime} \\ W &= \frac{\cos \theta}{v} \frac{\mathcal{L}(x, x_s)}{\mathcal{L}(x, x_r)} e^{-i\delta\tau} = \text{weight function} \\ \delta\tau &= \text{phase shift due to caustics} \end{aligned}$$

Wavefront construction

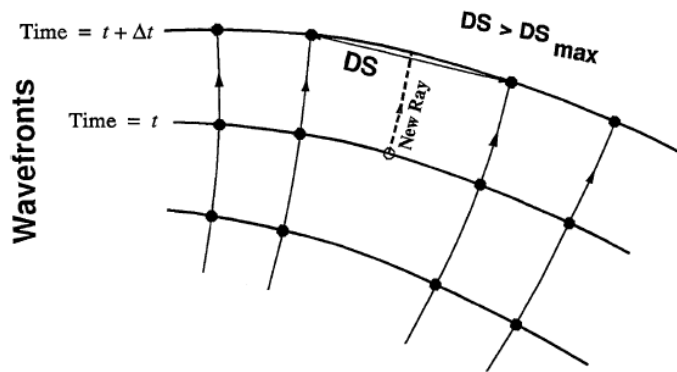
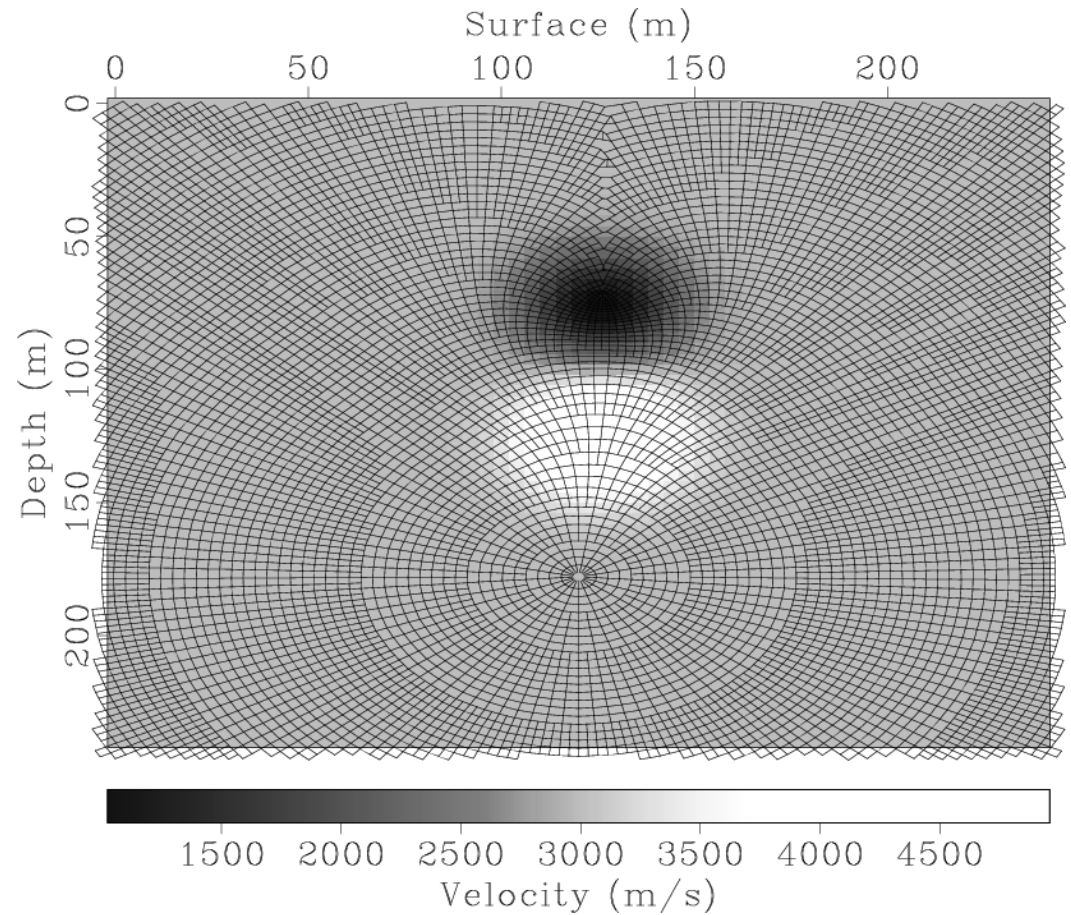


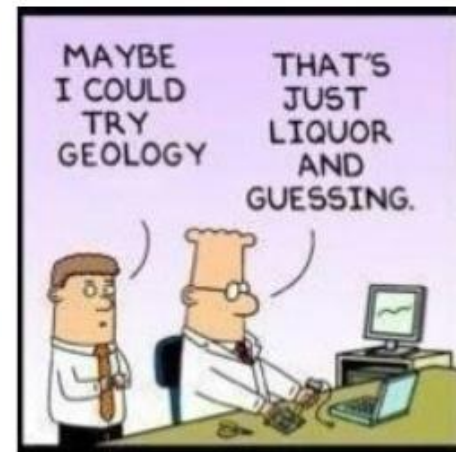
FIG. 6. Propagation of WFs and interpolation of a new ray in 2-D WF construction. When the difference between two rays becomes too large at the WF at time $t + \Delta t$, a new ray is interpolated on the WF at time t .

Vinje and Iversen, Geophysics (1999)



Applications of ray theory and ray tracing

- Applications of ray tracing:
 - Prestack depth migration (Kirchhoff family)
 - Seismic interpretation
 - Survey planning
 - **Generation of synthetic seismic data**
 - Theoretical investigations
- Different types of ray tracing:
 - Initial value ray tracing
 - Two-point ray tracing
 - Wave-front construction



Thermal modeling; heat rays

Fourier's law (steady-state temperature):

$$\mathbf{q} = -k \nabla T$$

T = temperature

k = thermal conductivity

$\mathbf{q}$ = heat flow

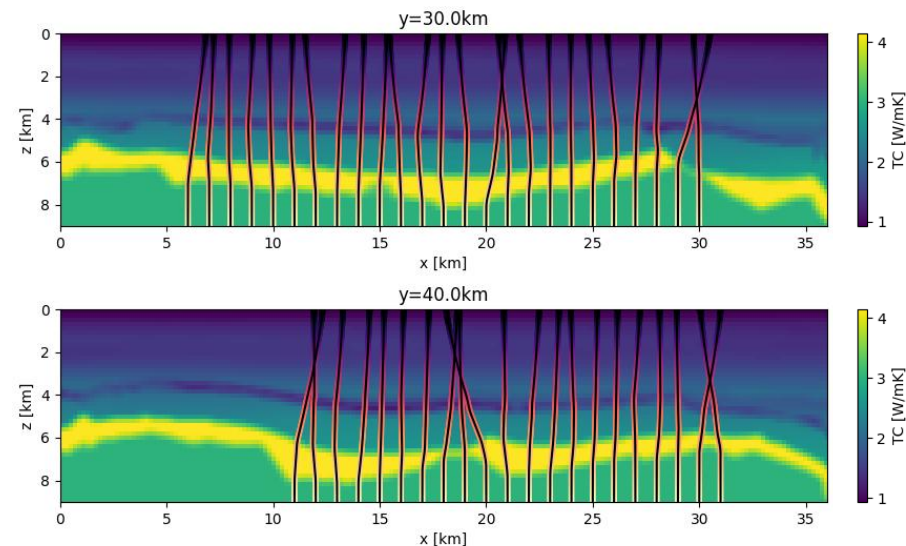
Eikonal equation for temperature:

$$\left(\frac{q}{k}\right)^2 = |\nabla T|^2$$

Can be solved with a seismic ray tracer if

$$\frac{k}{q} \rightarrow v$$

Method of characteristics



«Cold -flow» ray-tracing - isotherms

